Description Package Content Wiki

ESP32, ESP32

# ESP32-S3-Matrix Development Board, Onboard 8×8 RGB LED Matrix and QMI8658 Attitude Sensor, supports Wi-Fi and Bluetooth LE, ESP32 Development Board

| | |
|---|---|
| SKU | 27119 |
| Part No. | ESP32-S3-Matrix |
| Brand | Waveshare |

$8.99

1 ADD TO CART

| Price | Quantity |
|---|---|
| $8.99 | 1 |
| $8.59 | 2 |
| $8.39 | 3 |
| $8.31 | 4+ |

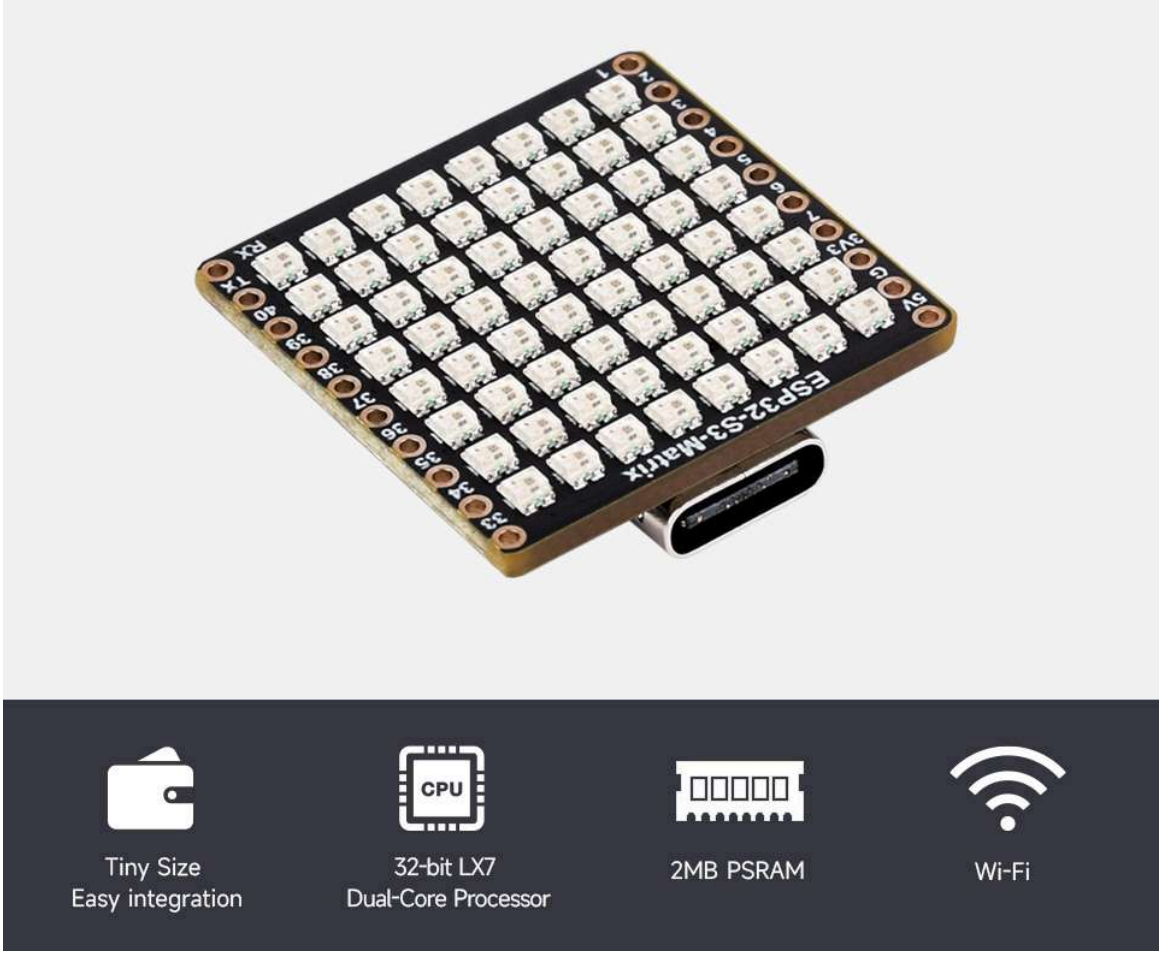
Contact

ESP32-S3-Matrix Development Board, Onboard 8×8 RGB LED Matrix And QMI8658 Attitude Sensor, Supports Wi-Fi And Bluetooth LE, ESP32 Development Board

# ESP32-S3 RGB LED Matrix Board

Based on ESP32-S3, 17 × multi-function GPIO pins

240 MHz Xtensa 32-Bit LX7 Dual-Core Processor




# Board Specifications

- Adopts ESP32-S3FH4R2 chip with Xtensa 32-bit LX7 dual-core processor, capable of running at 240 MHz
- Integrated 512KB SRAM, 384KB ROM, 2MB PSRAM, 4MB Flash memory
- Integrated 2.4GHz Wi-Fi and Bluetooth LE dual-mode wireless communication, with superior RF performance.

- Type-C connector, easier to use
- Onboard QMI8658 6-axis IMU (3-axis accelerometer and 3-axis gyroscope)
- Onboard 8 × 8 RGB LED matrix for colorful lighting display
- Adapting Dout pin for extending RGB matrix
- 17 × multi-function GPIO pins
- Rich peripheral interfaces for achieving various functions flexibly
- Supports multiple low-power operating states, adjustable balance between communication distance, data rate and power consumption to meet the power requirements of various application scenarios

# QMI8658 Gyroscope Sensor

Onboard QMI8658 6-Axis IMU For Tilt Angle And Speed Measurement

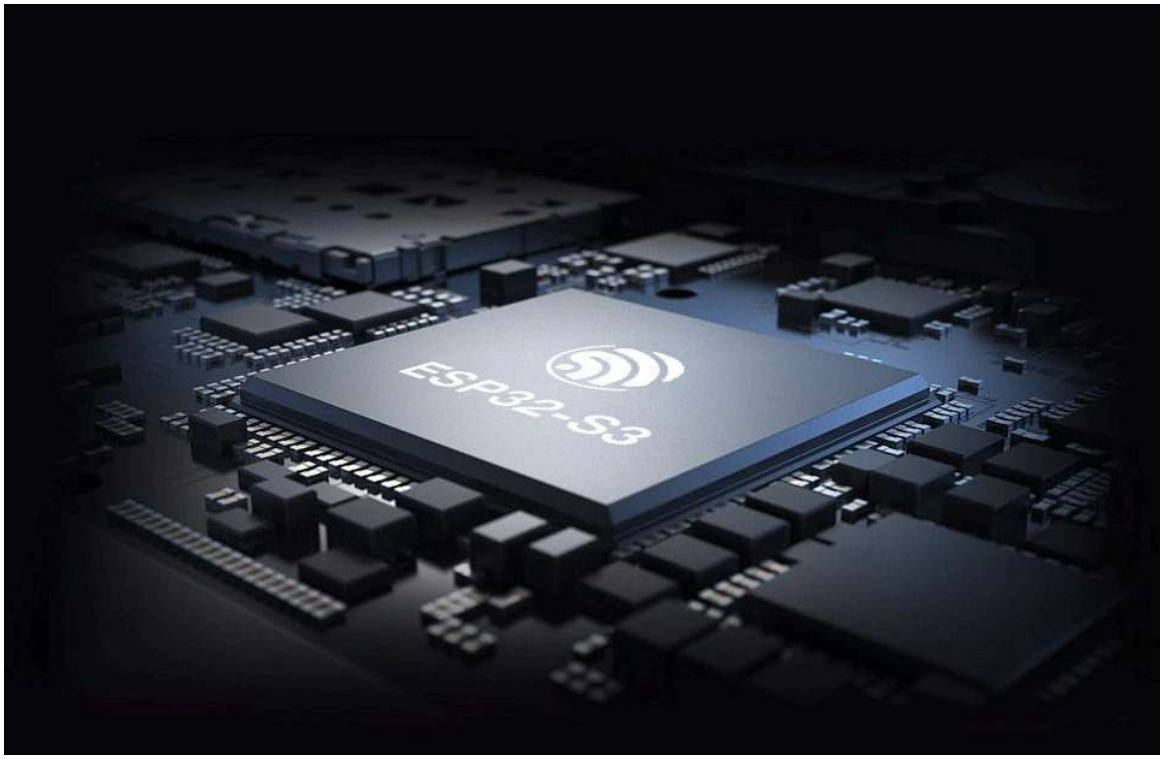
# Onboard 8×8 RGB LED Matrix

For Colorful Lighting Display

# Designed For AIoT Applications

Powered By Xtensa 32-Bit LX7 Dual-Core Processor With Frequency Up To 240 MHz

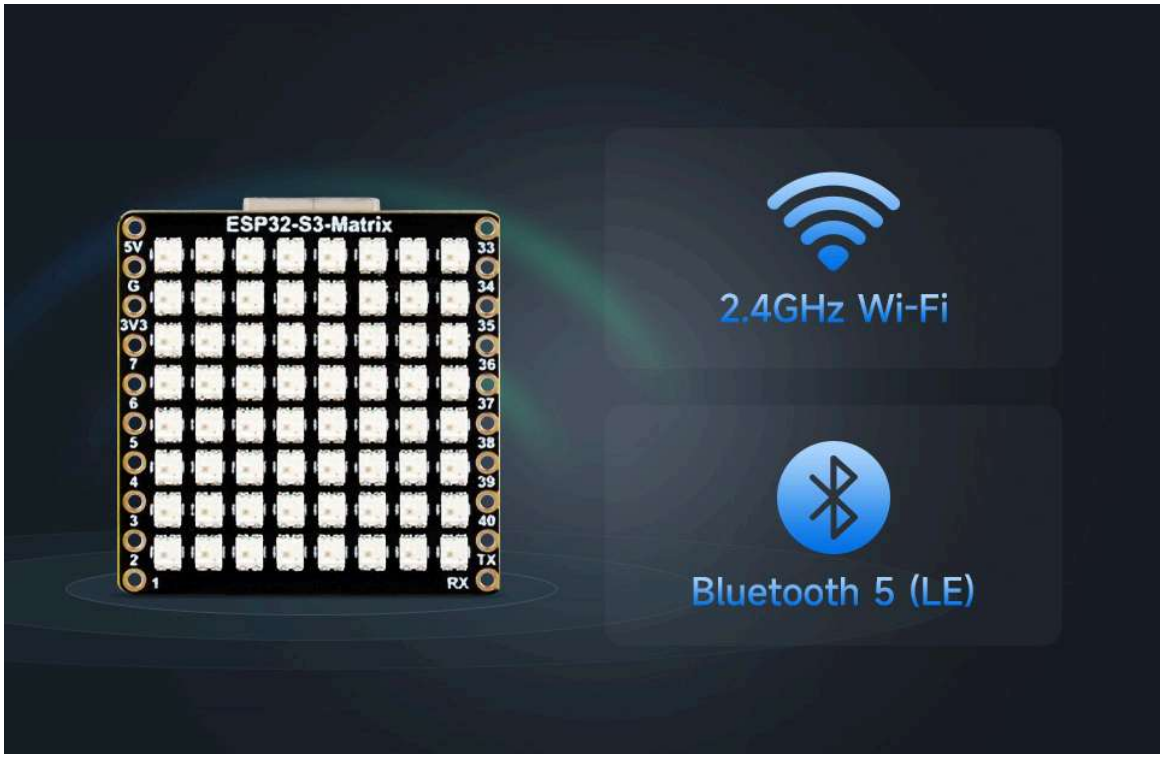
Powerful AI Computing Capability & Reliable Security Features



# WIFI And BLE 5 Support

ESP32-S3 Integrates 2.4 GHz Wi-Fi (802.11 B/G/N) With 40 MHz Of Bandwidth Support

Its Bluetooth Low Energy Subsystem Supports Bluetooth 5 (LE) And Bluetooth Mesh




# Supports ESP-IDF, Arduino, MicroPython

# Comprehensive SDK, Dev Resources, Tutorials To Help You Easily Get Started

## ESP-IDF

With free open source development tools, supports IDEs such as Eclipse and VSCode, easier for developers to use.

## Arduino IDE

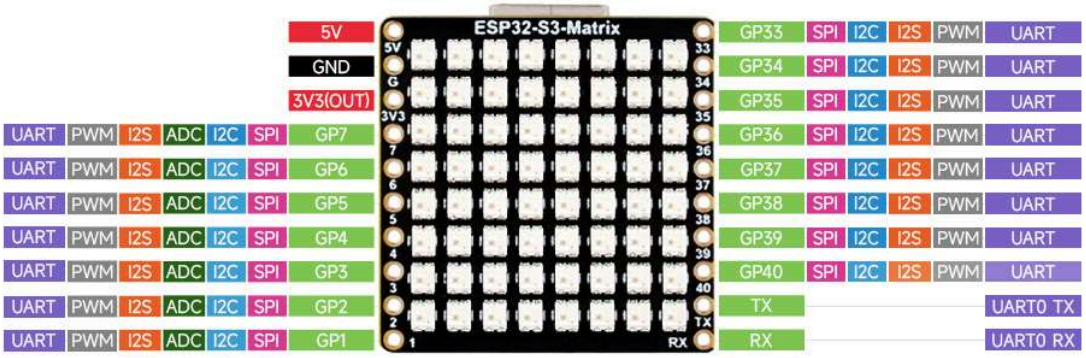
Arduino IDE is a open source electronic prototyping platform, convenient and flexible, easy to get started. After a simple learning, you can start to develop quickly.

## MicroPython

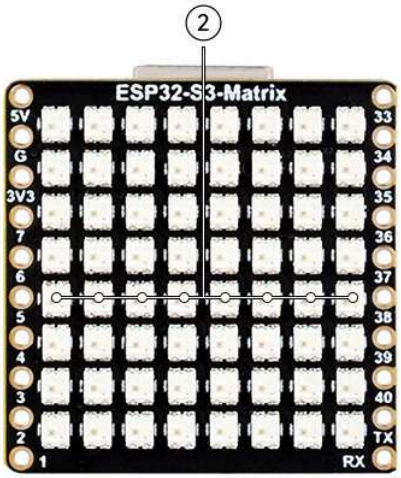
Micropython is a full implementation of the Python 3 programming language that runs directly on embedded hardware like Raspberry Pi Pico

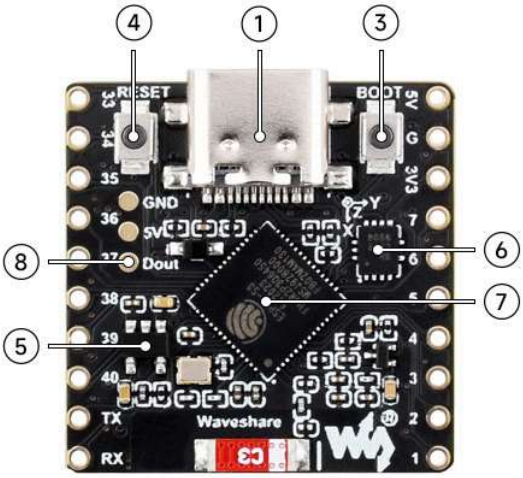
# 17 × Multi-Function GPIO Pins

## Configurable Pin Function, Allows Flexible Development And Integration




# What's On Board



1. **USB Type-C connector**
2. **8×8 RGB LED Matrix**
3. **BOOT button**

   press it when resetting to enter download mode
4. **RESET button**

5. **ME6217C33M5G**

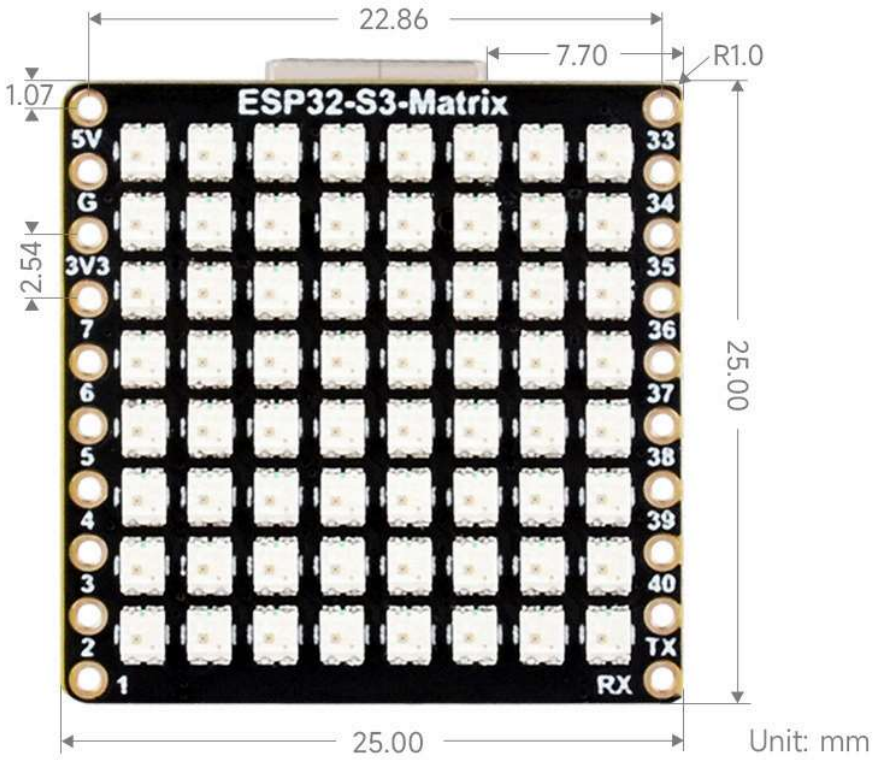
   Low dropout LDO, 800mA (Max)
6. **QMI8658**

   QST attitude sensor (3-axis accelerometer, 3-axis gyroscope)
7. **ESP32-S3**

   dual-core processor, up to 240MHz operating frequency
8. **Dout pin**

   for extending RGB matrix

# Outline Dimensions



# Resources & Services

WAVESHARE ORIGINAL

PROFESSIONAL TECH SUPPORT

RICH RESOURCES MANUAL & DEMO

**Wiki: www.waveshare.com/wiki/ESP32-S3-Matrix**